ABHIJEET

PhD Researcher | Reinforcement Learning | Optimal Control | Robotics

College Station, TX | abhijeetnirmal007@gmail.com | Website | GitHub | LinkedIn

Research Focus

Reinforcement Learning | Safe Learning | Trajectory Optimization | Numerical Optimization | Nonlinear Control | Robotics & Autonomous Systems

Education

Texas A&M University — College Station, TX

PhD in Aerospace Engineering, GPA: 4.0/4.0 — *Aug 2023 – Present; Expected 12/27*

Indian Institute of Technology Kanpur — Kanpur, India

M.Tech. in Aerospace Engineering, GPA: 10.0/10.0 — *Jan 2021 – Dec 2022*

Indian Institute of Technology Kanpur — Kanpur, India

B.Tech. in Aerospace Engineering, GPA: 8.2/10.0 — *Jul 2017 – Dec 2020*

Research Experience

Texas A&M University — College Station, TX

Graduate Research Assistant — *Aug 2023 – Present*

- Developed **Box-iLQR**, a constrained nonlinear trajectory-optimization framework combining iterative LQR with logarithmic barriers to compute feasible trajectories together with time-varying feedback policies.
- Developed an **SQP-based theoretical perspective on iLQR and DDP**, establishing descent properties for iLQR and analyzing their differing convergence behavior; presented at the **American Control Conference 2026**.
- Developed perturbation-feedback solvers for finite- and infinite-horizon nonlinear optimal-control problems, including closed-form feedback constructions, and benchmarked them against nonlinear-programming and receding-horizon approaches.
- Implemented **Sequential Quadratic Programming** and **Interior-Point** baselines for nonlinear trajectory optimization and applied optimization/feedback methods to spacecraft attitude and orbit-control problems under constraints and disturbances.

Texas A&M University — College Station, TX

Learning-Based Control Research — *Jun 2025 – Present*

- Implemented safe reinforcement-learning controllers using **PPO** and **TD3** with **CBF-QP safety filtering**, and benchmarked them against **SAC** on nonlinear unicycle navigation tasks.
- Evaluated policies across **15 independent training runs** and on unseen obstacle configurations to study reward, computational effort, safety, and generalization beyond the training environment.

Open-Source Research Software

Box-iLQR-Python — 2026 – Present

Project page / documentation — *Python*

- Built an actively developed Python implementation with state/control bounds, RK4/Euler discretization, analytical barrier derivatives, regularized backward passes, feasibility-preserving line search, continuation/warm starts, finite-difference derivatives, convergence diagnostics, and `pytest`-based automated tests.
- Structured the solver and examples for reuse by researchers studying constrained nonlinear optimal control and feedback trajectory optimization.

Prior Research Experience

Indian Institute of Science

Bengaluru, India

Aircraft Controls Researcher

May 2023 – Jul 2023

- Designed gain-scheduled control strategies for nonlinear aircraft dynamics, identified equilibrium regimes, and implemented linearized stabilization methods in Simulink across multiple flight conditions.

Earlier Robotics Experience

Space Robotics Lab, Indian Institute of Technology Kanpur

Kanpur, India

Spacecraft robotics and controls research

2021 – 2022

- Designed and fabricated an air-cushion hemispherical test platform and a **three-reaction-wheel actuation system** for three-axis spacecraft attitude-control experiments.
- Integrated Arduino-based hardware with **ROS** and Simulink for actuator commands and control-law testing, and developed a mobile platform for docking experiments.

Selected Publications

- **A Sequential Quadratic Programming Perspective on Optimal Control**
  American Control Conference (ACC), 2026.
  SQP interpretation of trajectory optimization with descent analysis and convergence comparisons for iLQR and DDP.
- **An Efficient iLQR Algorithm for Optimal Control with Bounded States and Inputs**
  AAS Astrodynamics Specialist Conference, 2026.
  Barrier-constrained iLQR for nonlinear trajectory optimization with state/control bounds and feedback policies.
- **Refining GLSDC for Orbit Determination**
  AAS Astrodynamics Specialist Conference, 2026.
  Angles-only orbit determination from short-arc noisy observations with analysis of multiple feasible solution classes.
- **An Optimal Solution to Infinite Horizon Nonholonomic and Discounted Nonlinear Control Problems**
  American Control Conference (ACC), 2025.
  Perturbation-feedback formulation for infinite-horizon nonlinear optimal-control problems.
- **Convexity in Optimal Control Problems**
  IEEE Conference on Decision and Control (CDC), 2024.
  Analysis of convexity structure and solution properties in nonlinear optimal-control formulations.

Technical Skills

Programming: Python, C, C++, MATLAB, Git
ML / Scientific Computing: PyTorch, NumPy, SciPy, `pytest`
Optimization & Control: iLQR, DDP, SQP, MPC, Control Barrier Functions, nonlinear optimization, trajectory optimization
Robotics & Simulation: MuJoCo, ROS / ROS 2, CasADi, Simulink

Selected Honors

- **Aero Graduate Excellence Fellowship**, Texas A&M University — 2024, 2025.
- **Aero Graduate Travel Award**, Texas A&M University — 2024, 2026.
- Member, **Tau Beta Pi**, Texas A&M University — 2024.

Additional Selected Research

- **Optimal Control with Lyapunov Stability for Space Applications** — AAS/AIAA Astrodynamics Specialist Conference, 2024; developed feedback-based optimal-control methods for robust spacecraft applications.
- **Control Strategy for Trading Off Solar Power and Control Input** — *Journal of Aerospace Engineering*, 2024; developed constrained control strategies balancing actuation effort and available solar power.